Late Paleozoic Remagnetization and Its Carrier in the Trenton and Black River Carbonates from the Michigan Basin[1]

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ABSTRACT

Paleomagnetic study of subsurface samples of the Upper Ordovician Trenton and Black River Limestones collected from six industrial wells in the southern part of the Michigan Basin yields a well-clustered characteristic direction (mean Dec/Inc of 170.6°/−10.9°, α_{95} = 7.6°) after rotation of random declinations by aligning a present day field overprint with geographic north. The corresponding paleomagnetic pole at 52.4°N 111.0°E is similar to those of the remagnetized carbonates in Ontario and New York State, suggesting that the Michigan formations were also remagnetized in the late Paleozoic. Rock magnetic characterizations suggest that the characteristic magnetization is carried by single and pseudosingle domain magnetite grains. Very fine rounded single crystal magnetite grains, believed to be products of chemical precipitation, observed by scanning and scanning transmission electron microscope (SEM/STEM) are inferred to be the carrier of the remagnetization. Paired carbon isotope analyses of bulk carbonate and anhysteretic susceptibility (X_{arm}) of cuttings from two wells were examined to test the hypothesis that oxidation of hydrocarbons may lead to formation of authigenic magnetite. Oil saturated horizons and overlying strata in a producing well exhibit lower X_{arm} intensities than the equivalent horizons in an adjacent non-producing well. Little correlation between X_{arm} and $\delta^{13}C$ is observed in carbonate-rich lithologies due to the masking of the hydrocarbon related cement compositions by the host carbonate. In contrast, carbonate-poor shale horizons overlying hydrocarbon-rich strata in the producing well exhibit strong correlative peaks in X_{arm} and $\delta^{13}C$. The overall lower X_{arm} intensities in the hydrocarbon-producing well suggest that it is not merely the presence of hydrocarbons which promotes authigenic magnetite formation. Rather, the magnetite/hydrocarbon relationship more likely records processes associated with hydrocarbon migration and interaction with oxidizing subsurface fluids in areas peripheral to sites of accumulation.

Introduction

A late Paleozoic remagnetization is recognized in many Paleozoic carbonates in the northeastern USA. Early/middle Paleozoic limestones in New York State, such as the Ordovician Trenton Formation and limestones of the Devonian Onondaga and Helderberg Group, have been completely remagnetized during the Kiaman reversed Polarity Superchron at the same time as the late Paleozoic Alleghenian Orogeny (Kent 1979, 1985; McCabe et al. 1983, 1984; Jackson et al. 1988). This remagnetization extends well into the midcontinent as far as Minnesota, Iowa, and Missouri (Lu et al. 1990). There are a few exceptions, most notably in the Silurian Wabash limestone (Indiana, McCabe et al. 1985) and in the Ordovician Oneota Dolomite along the Minnesota, Wisconsin, and Iowa border (Jackson and Van der Voo 1985).

In the Michigan Basin, carbonates of the Trenton and Black River Formations are known only from boreholes. The nearest outcrops of these units are in New York State, where the Trenton Formation is completely remagnetized (McCabe et al. 1984). Remagnetization of carbonates in the Michigan Basin is, therefore, a possibility that merits investigation, although the Trenton and Black River carbonates in the Michigan Basin represent localities quite far from the tectonic activity thought to have caused the remagnetization of the carbonates in New York State. Lu et al. (1990) reported a possible remagnetization of Upper Silurian dolomitic carbonate in the southeastern part of Michigan based on the strong anhysteretic remanent magnetization (ARM) intensity (16.5 mA/m).

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A second focus of this study is evaluation of the role of hydrocarbons in the authigenesis of magnetite. Elmore et al. (1987) and McCabe et al. (1987) have reported that the precipitation of authigenic (secondary) magnetite is possibly related to the presence/migration of hydrocarbons, although the exact relationship between hydrocarbons and secondary magnetite is controversial (Fishman et al. 1989; Elmore and Leach 1990). The Trenton and Black River Formations are among the largest hydrocarbon reservoirs in the Michigan Basin and adjacent regions. Budai and Wilson (1991) found that these formations have been diagenetically altered, and that the mineralization and hydrocarbon migration were related to compressional deformation caused by late Paleozoic Appalachian tectonism. Samples from the Trenton and Black River Formations are therefore ideal for testing possible remagnetizations of Permo-Carboniferous age and the controversial relationship, if any, between hydrocarbons and magnetite authigenesis.

This study is a part of the Michigan Gas Project for which geochemical and isotopic analyses of seepage gases, groundwater, wall rocks and overburden as well as paleomagnetic and rock magnetic measurements of wall rocks were employed to explore new methods of detecting the presence of gas and/or oil in the Michigan Basin. In this paper, we present paleomagnetic and rock magnetic measurements and SEM/STEM observations on subsurface samples of the Trenton and Black River Formations, and anhysteretic susceptibility (X_{arm}) and carbon isotope measurements on drill cuttings of strata of Carboniferous to Ordovician age from industrial wells in the Michigan Basin. These data contribute to an understanding of the process of remagnetization and the relationship between magnetite authigenesis and hydrocarbon distribution.

Sampling and Well Description

The Michigan Basin is a simple, undeformed and isolated intracratonic basin which covers the entire lower peninsula of Michigan, parts of Ontario, Ohio, and Indiana. It is separated from the Appalachian Basin by the Cincinnati, Findley and Algonquin arches and from the Illinois Basin by the Kankakee arch. At the center of the Michigan Basin, 5 km of Phanerozoic sedimentary rocks range in age from Cambrian to Quaternary (Fisher et al. 1988). Several petroleum companies have drilled wells in the southern part of the Michigan Basin in the Albion-Scipio and Stoney Point fields (figure 1). Five wells, Conklin 1-31, Casler 1-30, Casler 5-30, Tolle 1-33, and Hall 1-13, have penetrated the Trenton and Black River Formations and have provided cores for our paleomagnetic, rock magnetic, and SEM/STEM investigations (figure 1). In addition, two wells, Conklin 1-31 and Ester Weir 1-5 (figure 1), have provided drill cuttings (rock chips) of the lower Carboniferous Coldwater Shale through the Ordovician Black River carbonates every 10 to 20 feet. These cuttings have been used for anhysteretic susceptibility (X_{arm}) and carbon isotope measurements.

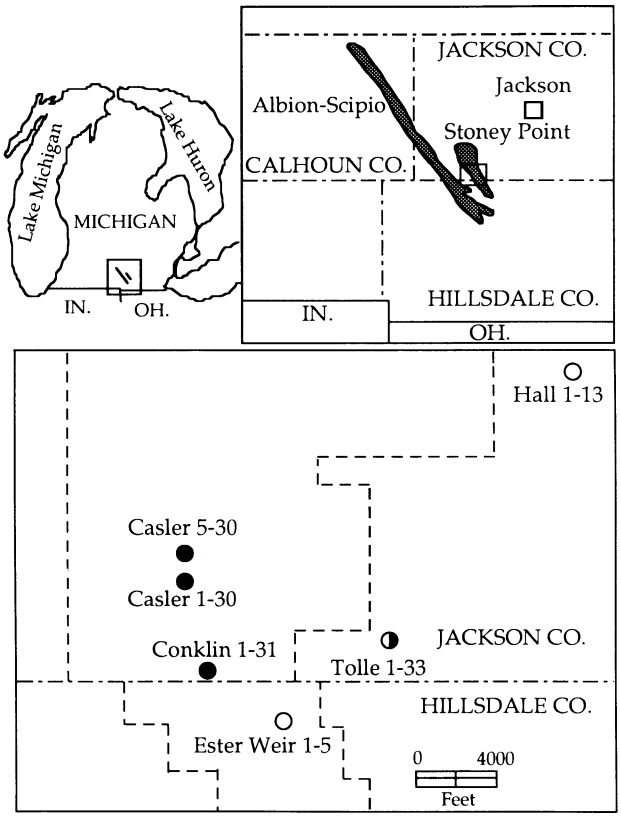


Figure 1. Location maps of the Albion-Scipio and Stoney Point field and five industrial wells studied in the southern part of Michigan Basin. Boxed areas are enlarged from top left figure in a clockwise manner. Symbols are solid circle, producing well; half-solid circle, producing but non-profitable well; open circle, dry well; dashed line, boundary of oil/gas field; and dash-dotted line, county boundary.

Conklin 1-31, Casler 1-30, and Casler 5-30 are producing wells and Tolle 1-33 is a producing but non-profitable oil well, while Hall 1-13 and Ester Weir 1-5 are dry wells (figure 1). Samples for paleomagnetic study were recovered from the depth of 3700 to 4360 feet. Figure 2 shows sampling depths and simplified lithologies for the five wells, from which a total of 116 samples from the Trenton and

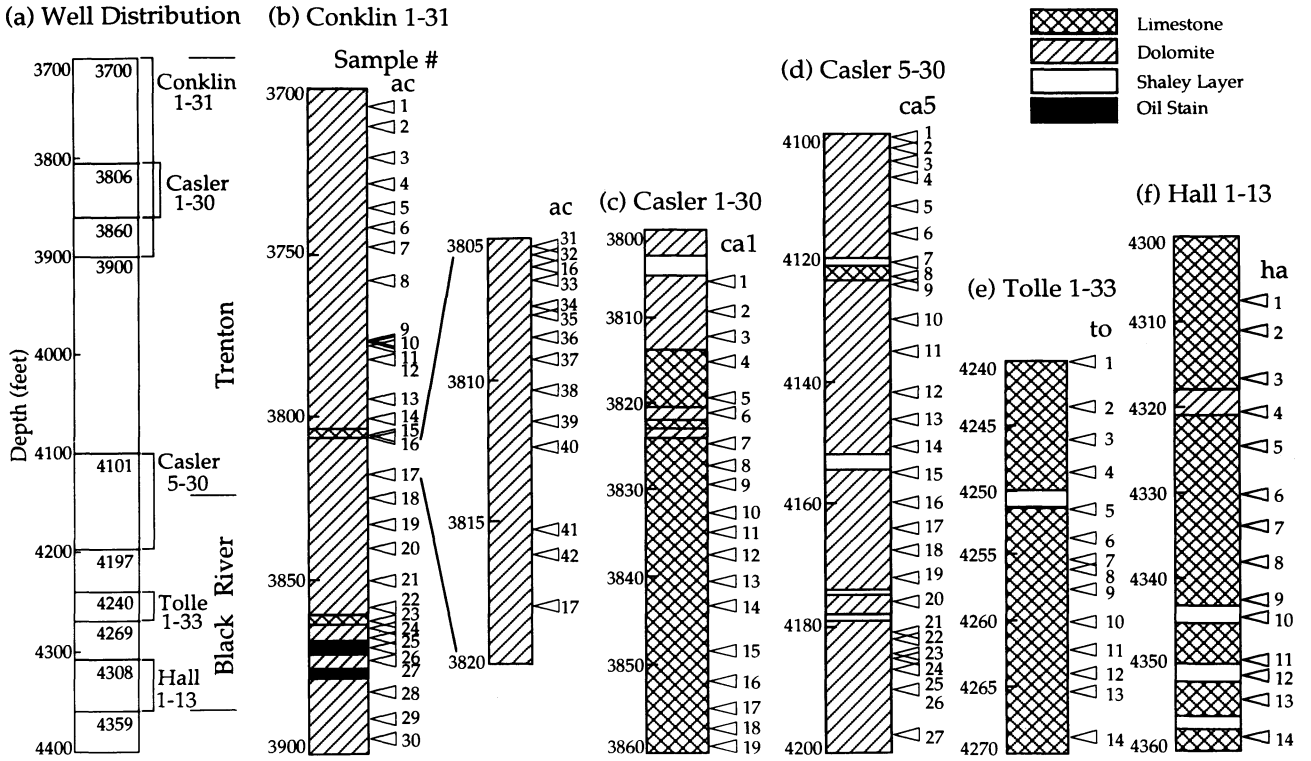


Figure 2. Sample distribution and simplified lithology of wells. (*a*) Depth intervals of Conklin 1-31, Casler 1-30, Casler 5-30, Tolle 1-33 and Hall 1-13 wells and corresponding formation names. (*b*)–(*f*) Sample distribution and simplified lithology with depth for Conklin 1-31, Casler 1-30, Casler 5-30, Tolle 1-33, and Hall 1-13, respectively.

Black River Formations were collected (figure 2*a*). Carbonate samples of the Trenton Formation were taken from Conklin 1-31, Casler 1-30, and part of Casler 5-30. Samples of the Black River Formation were taken from a lower part of Casler 5-30, Tolle 1-33 and Hall 1-13. Conklin 1-31 consists mainly of dolomite with a few thin limestone layers, and heavy oil stains at the bottom of the well (figure 2*b*). Casler 1-30, overlapping with Conklin 1-31 in depth, is comprised dominantly of limestone, minor dolomite, and shaley carbonate (figure 2c). In Casler 5-30, dolomite is a major phase with minor limestone and shaley beds. Samples #1 to #13 were taken from the Trenton Formation, while the rest of samples (#14 ~ #27) were taken from the Black River Formation (figure 2*d*). Tolle 1-33 and Hall 1-13 samples are predominantly limestones of the Black River Formation, which contains a thin dolomite layer and a few shaley layers (figure 2*e*, *f*).

Standard paleomagnetic samples (2.5 cm × 2.5 cm) were drilled perpendicular to the long axis of the well cores from the 5 inch diameter subsurface core material. Because the well cores were drilled perpendicular to bedding, the axes of the paleomagnetic specimens are aligned parallel to the bedding planes of sedimentary strata (figure 3). Given that the well cores are not azimuthally oriented, declinations in the paleomagnetic samples cannot be determined; on the other hand, these cores are vertically marked up/down, so that the inclinations can be determined directly.

Methods

For the paleomagnetic directional study, a two-axis ScT cryogenic magnetometer was used to measure the natural remanent magnetizations (NRM) in a magnetically shielded room at the paleomagnetic laboratory of the University of Michigan. Samples were subjected to stepwise thermal or alternating field (AF) demagnetization with Schonstedt equipment. Samples were thermally treated at 20 different steps, whereas AF demagnetizations were carried out at 5–10 mT intervals up to a peak field of 100 mT.

The intensity of characteristic remanent magnetization, anhysteretic susceptibility (X_{arm}, King et al. 1982) and low-field susceptibility (X) were measured for all paleomagnetic specimens. In addition, a modified Lowrie-Fuller test (Lowrie and Fuller 1971; Johnson et al. 1975), differentiated anhyster-

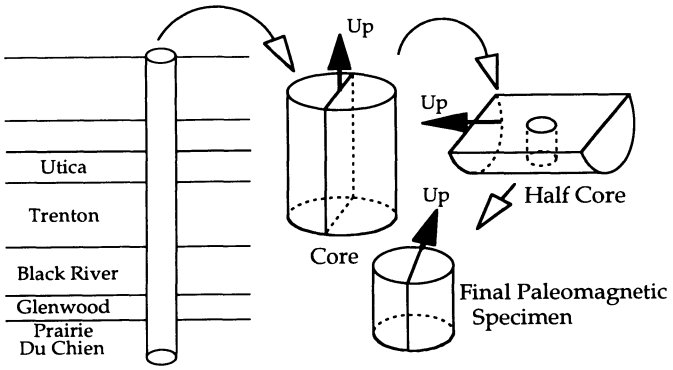


Figure 3. Sampling method for paleomagnetic specimens. Five inch well cores are drilled perpendicular to bedding plane, then paleomagnetic specimens are drilled perpendicular to the axis of the five inch core.

etic remanent magnetization (ARM, e.g., Dunlop 1972) or partial ARM acquisition (pARM, Jackson et al. 1988) and acquisition of saturation isothermal remanent magnetization (SIRM) were measured for selected specimens. The ARM was induced using a 100 mT AF field with a 0.1 mT DC biasing field, and isothermal remanent magnetization (IRM) was acquired using a 100 mT DC field for the modified Lowrie-Fuller test. These laboratory-induced ARMs and IRMs were subsequently AF demagnetized. The SIRM acquisition curves were generated by stepwise magnetization of specimens in 12 steps up to a maximum field of 1.4 T. The differentiated ARM was obtained by plotting ratios of the change in ARM intensity and the corresponding increment of the alternating field (dARM/dAF), versus AF values. The pARM, equivalent to the differentiated ARM, was obtained using a pARM switch that turns the DC biasing field on and off at preset points for a given interval of AF field during the decay of the field from a maximum of 100 mT in the Schonstedt AF demagnetizer. Low field susceptibility was measured using a SI-2 susceptibility bridge manufactured by Sapphire Instruments.

Thin sections of the carbonates were prepared for scanning electron microscope (SEM) observations using "sticky wax" as an adhesive, whereupon they were polished with 0.3 micrometer alumina powder. The SEM used is a Hitachi S-570, equipped with a Kevex Quantum energy dispersive spectroscopic (EDS) system for X-ray analysis, and a back-scattered electron (BSE) detector. Iron oxides were detected in BSE mode as it gives image contrast as a function of atomic number. Magnetic extracts were recovered from insoluble residues by dissolving limestones in buffered acetic acid of pH 5 (McCabe et al. 1983). X-ray powder diffraction patterns from magnetic extracts were used to identify iron oxides.

Carbon and oxygen isotopic compositions of bulk rock carbonate were determined by homogenizing 1 gram of well cuttings. These powders were then heated to 380°C *in vacuo* to remove volatile organic contaminates. Analysis was performed with an automated reaction/extraction system coupled directly to the inlet of a MAT251 ratio gas mass spectrometer. Molecular mass ratios were corrected for the contribution of ^{17}O and converted to enrichments relative to the PDB carbonate standard. External analytical precision, which is better than 0.1 per mil for both carbon and oxygen, was monitored through daily analysis of National Bureau of Standard (NBS) powdered carbonate standards.

Results

Paleomagnetic Results and Analyses. Natural remanent magnetization (NRM) intensities typically range between 0.1 and 7 mA/m. Demagnetization behavior is relatively simple (figure 4). Two components of magnetization are observed using orthogonal demagnetization diagrams (Zijderveld 1967). The first component has random declinations owing to the lack of azimuthal core orientation, but its inclinations conform to that of the present day field. The second component also yields random declinations, but has shallow negative inclinations suggesting an ancient magnetization. The second component is stable above 325°C up to the maximum unblocking temperature of 500°C in thermal demagnetization and has coercivities ranging from about 30 mT to 100 mT or higher in AF demagnetization. The intensities of the second (ancient) component range from 0.05 to 3 mA/m.

Although most of the specimens behave well during thermal and AF demagnetization, thermal demagnetization is more effective in separating the high temperature component from low temperature component (figure 4*a*), as was also observed in surface Trenton samples from New York and Canada (McCabe et al. 1984). A representative AF demagnetization diagram is shown in figure 4*b*. Specimen directions were determined by principal component analysis (Kirschvink 1980) using three or more points in the demagnetization diagram to determine individual directions. The high temperature–higher coercivity component has random declinations in an equal area stereoplot but shows an otherwise remarkable distribution of shallow

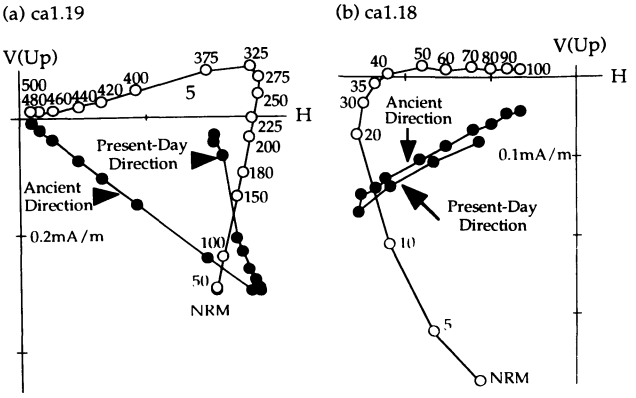


Figure 4. Representative orthogonal demagnetization diagrams (Zijderveld 1967) for the Trenton and Black River carbonates from the Michigan Basin showing bivectorial decay of the remanence during (*a*) thermal and (*b*) alternating field demagnetization. Open (solid) symbols represent projections onto a vertical (horizontal) plane.

upward inclinations (figure 5*a*). Although the declinations are random because the cores were not azimuthally oriented, we are fortunate in that the samples possess a low temperature–low coercivity component with a downward inclination of about 65° that can be safely assumed to be of recent origin (figure 4*a, b*). Thus we can align the declination of the overprints with geographic north and thereby determine the azimuthal orientation of the high temperature–higher coercivity component (e.g., Van der Voo and Watts 1978). Figure 5*b* shows the high temperature–higher coercivity components after this alignment has been performed.

The carbonates from the two different formations of Ordovician age yielded final directions (after rotation) remarkably similar to each other. The Fisherian mean direction of all specimens from five wells treated by thermal and AF demagnetizations is D = 170.5° and I = −10.9° (α_{95} = 7.6). This direction is statistically indistinguishable from separate mean directions for each well and to the means of samples treated by either thermal or AF demagnetization. The corresponding paleomagnetic pole (site location 42.17°N and 84.56°W) is Lat. = 52.4°N Long. = 111.0°E. When plotted on the Paleozoic and Lower Mesozoic segment of the North American apparent polar wander path (APWP, Van der Voo 1990), this pole falls between the Upper Permian and the Lower Triassic pole positions of North America (figure 5*c*). This paleomagnetic pole is clearly different from the Ordovician pole, indicating that the Trenton and Black River carbonates were remagnetized.

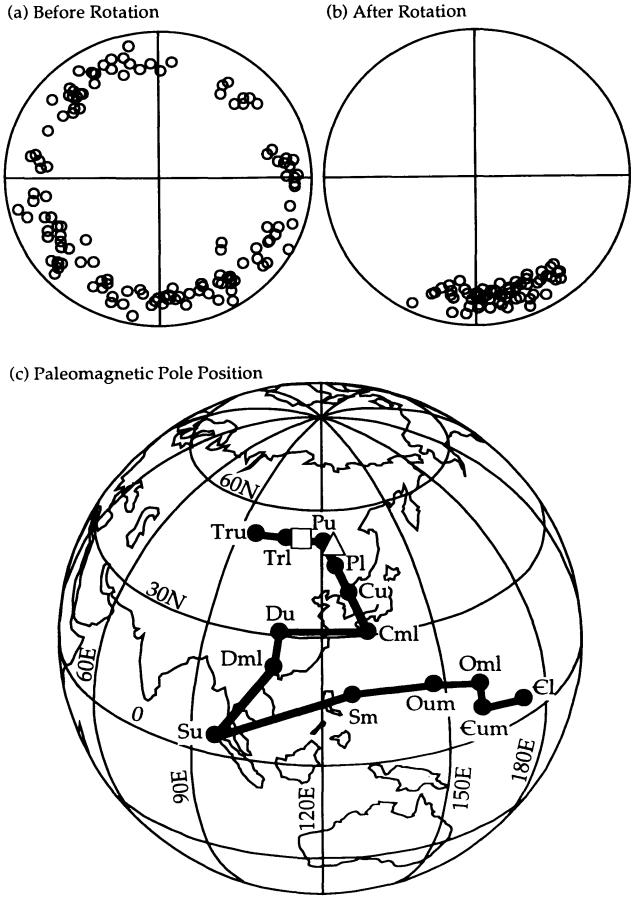


Figure 5. Equal area projections of ancient directions (*a*) before and (*b*) after rotation based on alignment of the present-day field direction. (*c*) Paleomagnetic pole position for the Trenton and Black River carbonates from the Michigan Basin (open square) and the mean paleomagnetic pole position of the remagnetized carbonates of early/middle Paleozoic age from New York state (open triangle, Scotese et al. 1982; McCabe et al. 1984; Kent 1985). The Paleozoic and the Lower Mesozoic portion of the apparent polar wander path for North America is after Van der Voo (1990). Symbols are: €l, lower Cambrian; €um, upper/middle Cambrian; Oml, middle/lower Ordovician; Oum, upper/middle Ordovician; Sm, middle Silurian; Su, upper Silurian; Dml, middle/lower Devonian; Du, upper Devonian; Cml, middle/lower Carboniferous; Cu, upper Carboniferous; Pl, lower Permian, Pu, upper Permian, Trl, lower Triassic; and Tru, upper Triassic.

Rock Magnetic Properties. The intensities of the characteristic remanent magnetization (referring to the high temperature–higher coercivity component), anhysteretic susceptibility (X_{arm}), and low field susceptibilities (X) were examined to determine if there are any systematic variations between these values as a function of depth or for

particular lithologic features. A modified Lowrie-Fuller test (stability of ARM and IRM against AF demagnetization) and acquisition of SIRM were used to identify magnetic mineralogy and magnetic domain state in the carbonates from the Michigan Basin.

X_{arm}, X, and intensity of characteristic remanent magnetization are plotted for each well against depth (figure 6*a–e*). The variations in the intensity of characteristic remanence (broken line) are very similar to those of X_{arm} (solid line) in all wells, while changes in these two values are different from those of low field susceptibility (X, dotted line). This relationship is most prominently displayed in the Conklin 1-31 well (figure 6*a*) in that the values of X_{arm} and the intensity of characteristic remanence gradually increase with depth, whereas the low field susceptibility decreases. However, the difference between X_{arm}/intensity of characteristic remanence and X, observed in Conklin 1-31, Casler 1-30, and Casler 5-30, becomes less prominent in Tolle 1-33 and Hall 1-13 (figure 6*a–e*). Even though there is no systematic variation in the magnetic properties with depth and lithologic features, two peaks coincide with lithologic features. One peak at 3863 feet in the Conklin 1-31 well (arrow 1 in figure 6*a*) corresponds to an area of heavy oil staining and high porosity. The peak observed at 4257 feet in the Tolle 1-33 well corresponds to an interval of high porosity.

Changes in X_{arm} associated with depth are matched by similar changes in intensity of characteristic remanence. These changes reflect a variation in the concentration of magnetic minerals. X_{arm} curves for all five wells are plotted in figure 6*f*. The producing wells (curves a, b, and c in figure 6*f*) do not necessarily have higher X_{arm} values (concentration of magnetic minerals) than those of the non-producing wells (curves d and e in figure 6*f*); nor does lithology (dolomite, limestone) appear to correlate with intensity.

The SIRM was imparted in a stepwise manner up to 1.4 T. All specimens saturate between 0.3 and 0.4 T (figure 7*a*). The SIRM acquisition curves indicate that the major magnetic carrier in these carbonates is magnetite. Figure 7*a* shows the SIRM curves of representative samples for each well. The value of SIRM intensity varies from well to well reflecting the changes in magnetite concentration.

Results of pARM (or differentiated ARM) are shown in figure 7*b*. Samples from producing wells, ac30.1, cal.6 and ca5.22a, have peak values at around 30 mT, and these correspond to magnetite grains of 2–3 micrometer size (Marvin et al. 1986). Samples from non-producing wells (to4a and ha12b) show a peak shifted toward lower fields (about 15 mT) suggesting 5-micrometer-sized magnetite grains (Marvin et al. 1986). However, persistent large tails in the high field portion of all the curves indicate the additional presence of fine-grained magnetite in the samples. The modified Lowrie-Fuller test reveals that IRM is more stable than ARM in AF demagnetization (figure 7*c, d*) and yields a multidomain (MD) test result. Although the stabilities of ARM and IRM against AF demagnetization are those for MD particles, the inflected shape of decay curves, rather than the configuration of ARM and IRM decay curves, suggests single-domain (SD) and pseudo-single domain (PSD) carriers of the remanence (Dunlop 1983) in addition to multi-domain particles.

X_{arm}-$\delta^{13}C$ Relationship. Since peaks of high magnetite concentration as inferred from the X_{arm} measurements were coincident with a section of heavy oil stain at the bottom of the Conklin 1-31 well (arrow 1 in figure 6*a*) and a high porosity area in the Tolle 1-33 well (arrow 2 in figure 6*d*), it was suggested that there may be a relationship between the presence of oil/gas and enhancement of magnetite authigenesis. In an attempt to test this relationship, additional cuttings from this oil-producing well (Conklin 1-31) and adjacent dry well (Ester Weir 1-5) were prepared for paired X_{arm} and carbon isotope ($\delta^{13}C$) measurements. The Conklin 1-31 and Ester Weir 1-5 provided sample chips ranging from the Lower Carboniferous Coldwater Shale to the Upper Ordovician Black River carbonate.

Our approach relies upon the large difference in $\delta^{13}C$ that exists between carbon derived from hydrocarbon sources relative to typical sedimentary carbonates to trace the source of diagenetic carbonate cement. For example, primary marine carbonates of Phanerozoic age define a narrow range in carbon isotopic composition from -1 to $+5$ per mil (PDB) (Lohmann 1987). Other studies (Taylor 1982; Budai and Wilson 1991) further suggest that $\delta^{13}C$ of diagenetic carbonates that formed in settings unrelated to hydrocarbon influences rarely exceed -8 per mil $\delta^{13}C$. In contrast, carbonate cements related to the oxidation of hydrocarbons (-28 to -80 per mil) are marked by excursions in the carbon isotopic composition to highly negative values. Therefore, the presence of carbon isotope anomalies with values more negative than -8 per mil likely reflect the production of bicarbonate from the breakdown of hydrocarbons. Carbonate cement formed from such interactions is distinctive in its isotopic composition; however, de-

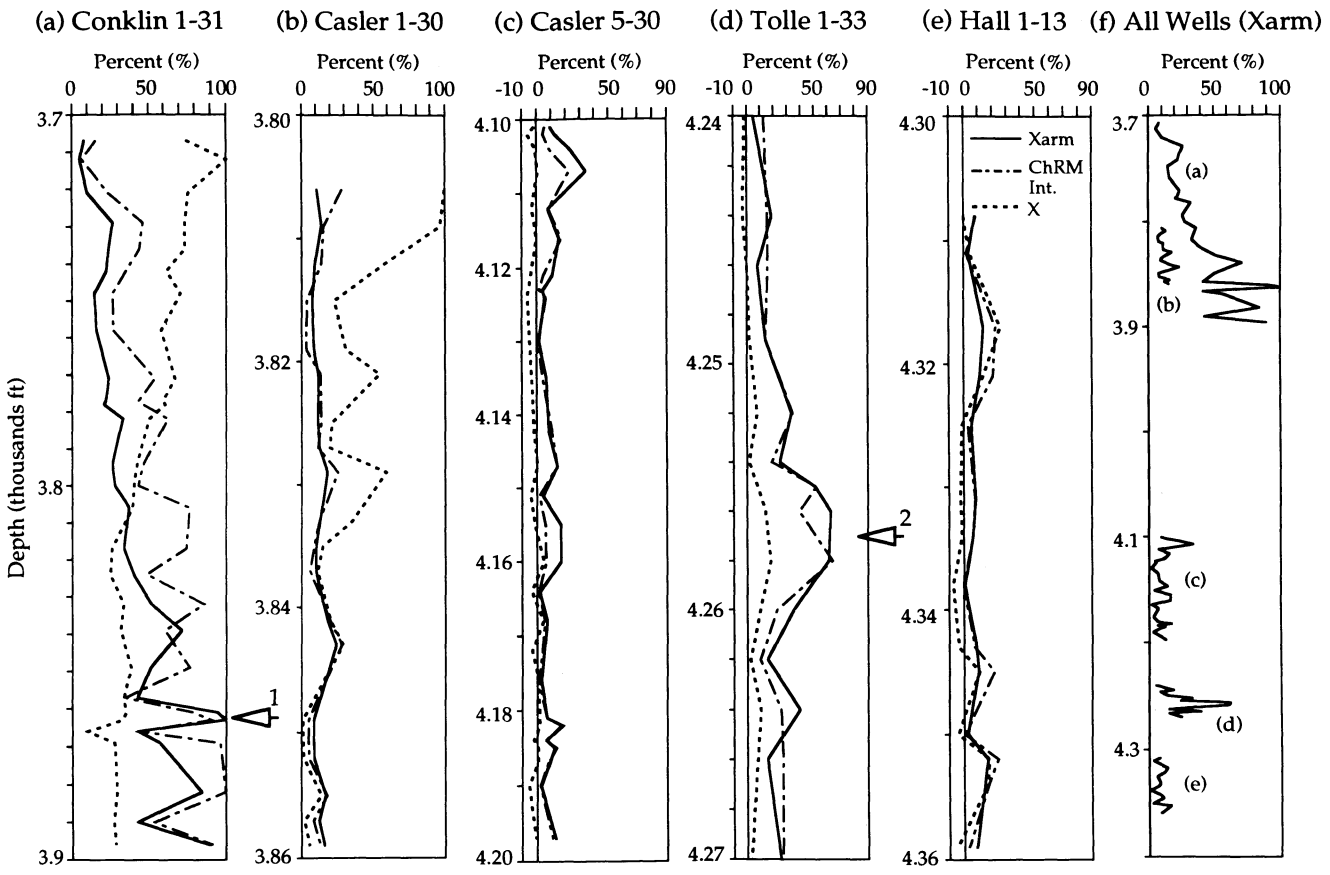


Figure 6. Normalized magnetic properties of samples from (*a*) Conklin 1-31, (*b*) Casler 1-30, (*c*) Casler 5-30, (*d*) Tolle 1-33, and (*e*) Hall 1-13. (*f*) Changes in the anhysteretic susceptibilities (X_{arm}) for the five wells are plotted. Symbols are: solid line, X_{arm}; dotted line, low field susceptibility (X), dash-dotted line, intensity of characteristic magnetization; arrow 1, area with heavy oil stain; and arrow 2, areas of high porosity.

pending upon its proportion relative to pre-existing carbonate in a sample, the magnitude of the anomaly may be significantly attenuated. For example, in carbonate-rich horizons such as the Trenton and Black River Formations, correlation between X_{arm} and $\delta^{13}C$ values may be masked by the contribution of the dominant primary carbonate. This contrasts with horizons lacking primary carbonates, such as shale or sandstones where diagenetic cement may comprise the bulk of the carbonate present within a single lithology. In such cases, a strong correlation should be present.

The $\delta^{13}C$ values for the Conklin 1-31 well range from +2.87 to −25.96 per mil (figure 8*a*), and those for the Ester Weir 1-5 well range from +2.40 to −28.93 per mil (figure 8*b*). However, the $\delta^{13}C$ values for the Trenton and Black River Formations, which are regionally producing horizons, vary from +0.15 to +1.30 per mil and from +1.34 to −0.93 per mil for the Conklin 1-31 and Ester Weir 1-5 wells, respectively. More negative $\delta^{13}C$ values are obtained from samples of shale layers such as the Coldwater, Sunbury, Antrim and Salina G-unit shales of the Salina Group (figure 8*a*, *b*). Samples from shaley intervals of the Traverse Formation and uppermost Dundee limestone in the Conklin 1-31 well have the most negative $\delta^{13}C$ values (−4.4 to −6.7 per mil, figure 8*a*) among carbonates. X_{arm} values of the cuttings vary between 0.09 and 59.55 mA/m for the Conklin 1-31 well (figure 8*a*), and the values for the Ester Weir 1-5 well range from 0.13 to 177.12 mA/m (figure 8*b*). Notably, X_{arm} intensities of the Trenton and the Black River carbonates are greater in the Ester Weir 1-5 well than in the Conklin 1-31 well. The X_{arm} values are plotted with the $\delta^{13}C$ values for the two wells for comparison (figure 8*a*, *b*). Several correlations between the X_{arm} and $\delta^{13}C$ values are observed. Three specimens of the Coldwater Shale from the Conklin 1-31 well at depths of 930, 1150, and 1170 feet show increase in the X_{arm} intensity and change toward more negative $\delta^{13}C$ values (arrows 1 and 2 in figure 8*a*). Although the $\delta^{13}C$ values are less negative than −8 per mil, a specimen from the Traverse

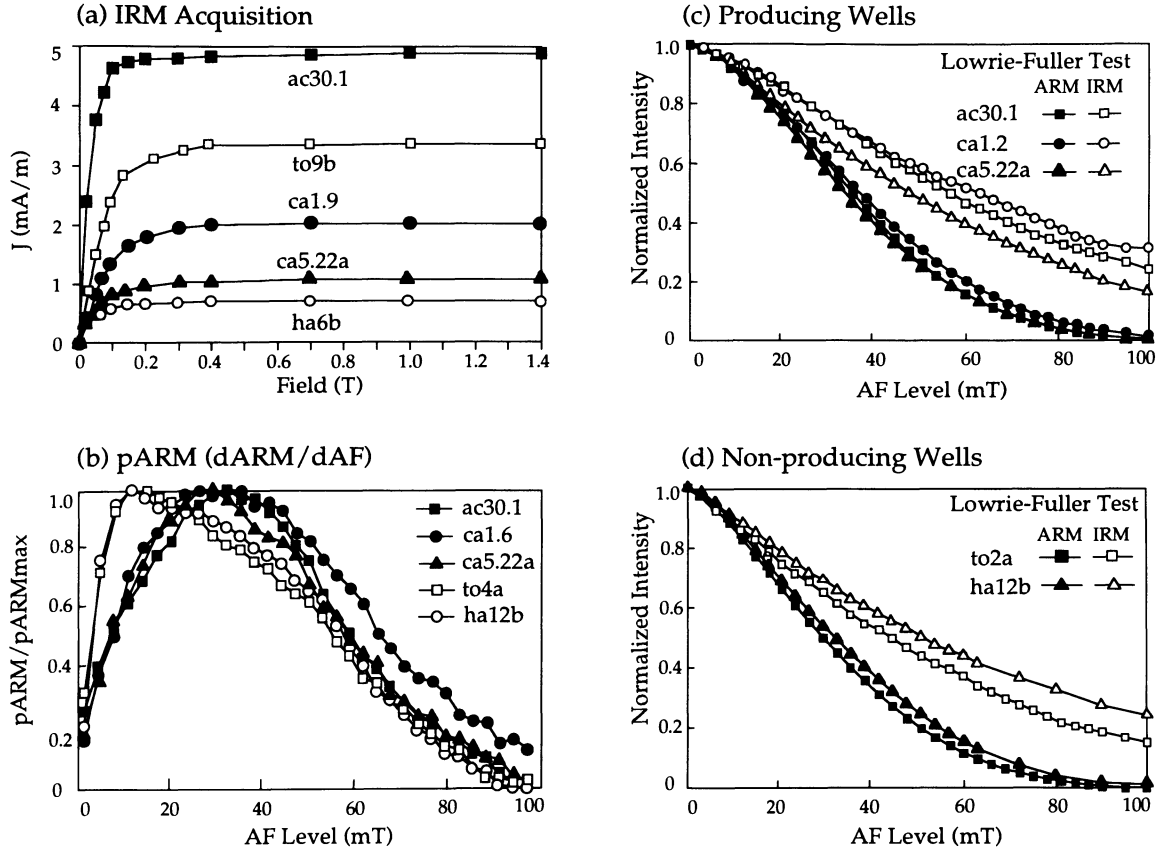


Figure 7. Rock magnetic properties of representative samples from the five wells. (*a*) Acquisition of isothermal remanent magnetization (IRM) approaches saturation in fields of less than 0.3 Tesla, indicating that magnetite is the dominant remanence carrier. (*b*) Partial anhysteretic remanent magnetization (pARM) curves showing that the most significant contribution to the remanence comes from 2–5 micrometer sized magnetite. (*c*), (*d*) Stabilities of IRM and ARM against alternating field demagnetization (modified Lowrie-Fuller test, Lowrie and Fuller 1971; Johnson et al. 1975) indicate multidomain (MD) magnetite behavior for (*c*) producing and (*d*) non-producing wells. Inflected decay curves of the remanences support single-domain (SD) and pseudo-single domain (PSD) test results.

Formation, one from the Dundee limestone, and a sample of the Utica Shale at depths of 1430, 1510, and 3430 feet, respectively, reveal correlations between the two values (arrows 3, 4, and 5 in figure 8*a*) in the Conklin 1-31 well. In the Ester Weir 1-5 well, X_{arm} intensitites in the carbonate and shale layers above the producing horizons are greatest and notably lower in the overlying Carboniferous Coldwater Shale without correlative $\delta^{13}C$ anomalies exept one positive correlation of the two values in a specimen of the Antrim Shale at 1395 feet (arrow 1 in fig. 8*b*). The carbonate-poor Coldwater Shale in the Conklin 1-31 well, however, exhibits increases in X_{arm} with correlative $\delta^{13}C$ anomalies (figure 8*a*) likely related to oxidation of methane escaping from the underlying hydrocarbon reservoirs in the Trenton and Black River Formations (e.g., Donovan et al. 1974).

SEM/STEM Observations. SEM observations of thin sections reveal considerable pyrite in the form of framboids (figure 9*a*), and euhedral to subhedral pyrite grains. Pyrite in the Michigan carbonates does not show any indication of the replacement by magnetite that was found in the New York carbonates (Suk et al. 1990). Typical submicrometer-sized rounded iron oxide grains (figure 9*b*) were observed in magnetic extracts. The uniform size and shape of the grains as well as their ideal composition (only Fe and O were detected) indicate that these grains are products of chemical precipitation in the carbonates. SEM observations of the magnetic extracts reveal that the submicrometer-sized grains are a dominant phase. These grains are usually clustered due to their magnetic attraction during sample preparation. The clustering of magnetic extracts made it difficult to obtain

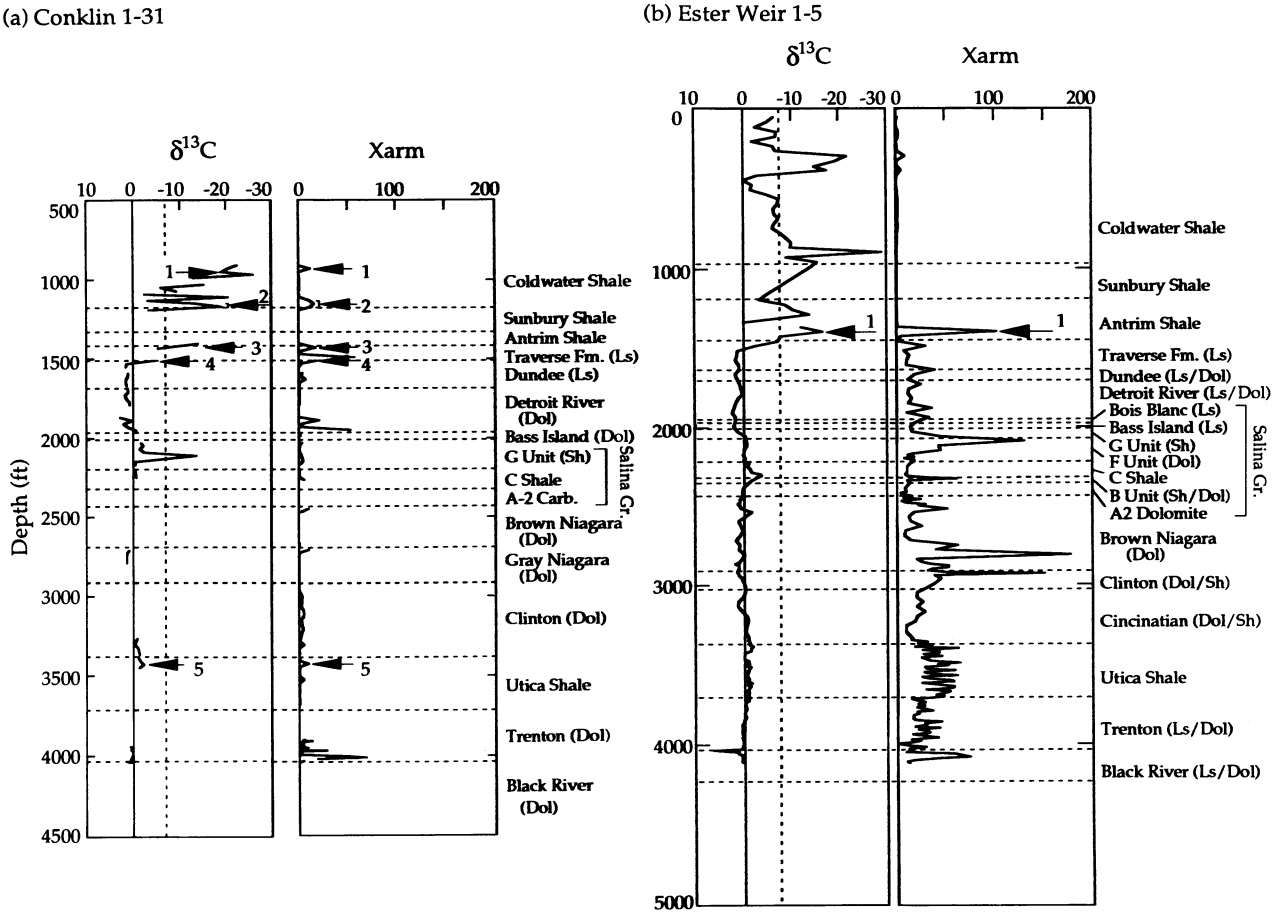


Figure 8. Carbon isotope ratios ($\delta^{13}C$) and anhysteretic susceptibilities (X_{arm}) of cuttings from (*a*) Conklin 1-31 and (*b*) Ester Weir 1-5 wells. Vertical dotted lines in carbon isotope diagram are drawn at −8 per mil. Arrows indicate samples having positive correlation between $\delta^{13}C$ and X_{arm} values. The Trenton and Black River formations are producing horizons.

selected area electron diffraction (SAED) in STEM observations. However, an isolated iron oxide grain of 0.1 micrometers (figure 9*c*) was shown to be single crystal magnetite from its SAED pattern (figure 9*d*). Similarities in size and shape between the isolated magnetite grain (figure 9*c*) and individual iron oxide grains of the clusters (figure 9*b*) imply that the clustered iron oxides are aggregates of SD- and PSD-sized magnetite grains.

Discussion

The paleomagnetic direction of the high temperature–higher coercivity component (D = 170.5°, I = −10.9°), measured in samples of the Trenton and Black River carbonates from the Michigan Basin is similar to directions found in the Ordovician Trenton limestone and Devonian Onondaga and Helderberg limestones in New York state that were remagnetized in the late Paleozoic (McCabe et al. 1984; Kent 1985; Scotese et al. 1982). The paleomagnetic pole is located at 52.4°N, 111.0°E (figure 5*c*). This pole falls between the Upper Permian and Lower Triassic pole positions of North America, but is clearly different from the Ordovician poles, thereby indicating a remagnetization. Although this paleopole from the Michigan carbonates is close to the paleopoles of the remagnetized carbonates of early/middle Paleozoic age from New York state (mean pole at 50.8°N, 123.5°E), the paleopole of the Ordovician carbonates in the Michigan Basin is closer to the Lower Triassic pole position of the North America than those from New York. While it may be possible that this difference in pole location is due to a different age, we cannot make this a firm conclusion because of the uncertainty introduced by the lack of azimuthal core orientation. A minor "rotation" of the Michigan mean direction brings it into good agreement with the paleopoles of the New York rocks.

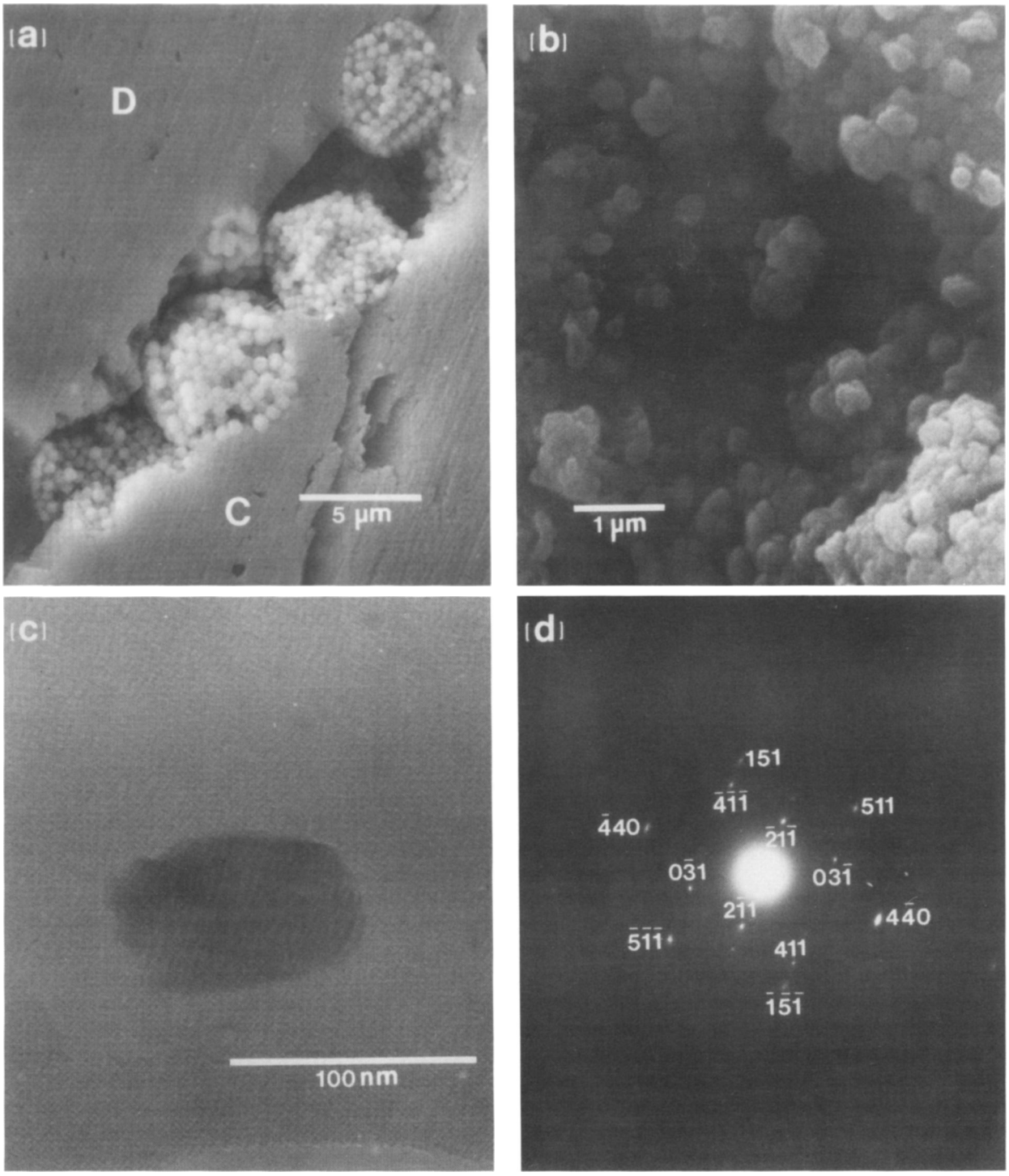


Figure 9. Scanning and scanning transmission electron microscope (SEM/STEM) images of representative iron oxides in the carbonates. (*a*) SEM image of framboidal pyrite along a grain boundary between dolomite (D) and calcite (C), secondary electron image (SEI). (*b*) SEM image of a cluster of submicrometer size iron oxide grains in magnetic extract, SEI. (*c*) Bright field image of a single crystal magnetite observed in STEM. (*d*) Single crystal magnetite selected area electron diffraction (SAED) pattern of the grain shown in (*c*).

Miller and Kent (1988) have argued that the ages of remagnetization are older in the southern Appalachians and younger to the north; it may well be that an east–west age difference exists as well, with the Michigan Basin, being farther from the Appalachian tectonism, experiencing a later remagnetization event than New York State. Be that as it may, the age of remagnetization of the Trenton and the Black River carbonates in the Michigan Basin is no younger than latest Paleozoic.

Howell and van der Pluijm (1990) observed that depositional facies architecture in the Michigan Basin reflects episodic subsidence patterns. Periods of episodic subsidence correlate temporally with orogenic events in the Appalachians for Cambrian through at least Silurian time. We have found that the age of the remagnetization in carbonate samples from the Michigan Basin correlates with the late Paleozoic Alleghenian Orogeny. Budai and Wilson (1991), noting that the Trenton and Black River Formations showed a complex diagenetic history, concluded that the mineralization and hydrocarbon migration in the basin are related to compressional deformation caused by the late Paleozoic tectonism of the Alleghenian Orogeny.

Unblocking temperatures of 500°C along with IRM acquisition experiments (figure 7*a*) for the Michigan carbonates indicate that magnetite is the principal magnetic carrier. The variation in the intensity of characteristic remanence is very similar to the variation of X_{arm}, whereas changes in these two values are different from changes in low field susceptibility (figure 6*a–e*). Based on the fact that X_{arm} is more sensitive to single domain (SD) and pseudo-single domain (PSD) behavior, whereas low field susceptibility is more sensitive to large multidomain grains (King et al. 1982), we believe that the characteristic remanent magnetization in these carbonates is carried by SD and PSD magnetite.

Our pARM (or differentiated ARM) experiments suggest that there is a significant proportion of grains of high coercivity in the Michigan carbonates. This is evidenced by the persistent large tails in the high coercivity spectra (figure 7*b*). The 2–5 μm magnetite size fraction was inferred to make the largest contribution to the pARM curves (fig. 7*b*) on the basis of a comparison with the results of Marvin et al. (1986). However, that size may be overestimated because Marvin et al. (1986) used highly stressed crushed magnetite whose grain size is as much as one order larger than that of hydrothermally precipitated stress-free magnetite for the same measured coercivity (Heider et al. 1987). If this is the case, the pARM results indicate abundant submicron-sized chemically precipitated magnetite, supposedly SD or PSD particles, in the Michigan carbonates. The modified Lowrie-Fuller test yields contradictory results because the IRM is more stable against AF demagnetization than ARM, indicating a predominance of multidomain (MD) grains (figure 7*c, d*). This was also observed for the Trenton limestone of New York State (McElhinny and Opdyke 1973). Dunlop (1983), however, concluded that the Lowrie-Fuller test is not capable of fine-scale classifications of domain structure or grain sizes and that the Lowrie-Fuller characteristics are simply an expression of the shape of remanence decay curves. The decay curves of the Michigan carbonates have the inflected shape characteristic of SD or PSD particles, as suggested by Dunlop (1983). Jackson (1990) found evidence supporting Dunlop's (1983) conclusion in a rock magnetic investigation on the Knox dolomite (Tennessee). The Lowrie-Fuller test for the Knox dolomite indicated SD-PSD behavior whereas almost 40% of the intensity of room-temperature SIRM of the Knox dolomite was removed after cycling to liquid nitrogen temperature in zero-field (low temperature demagnetization), indicating MD behavior (Dunlop and Argyle 1989). We conclude that, although the Lowrie-Fuller test indicates MD behavior in the Michigan Basin carbonates, more sensitive rock magnetic experiments clearly indicate a preponderance of SD and PSD grains. The existence of SD-PSD size grains is supported by SEM/STEM observations of magnetic extracts (figure 9*b–d*). These fine magnetite grains in the range of the SD and PSD state are inferred to be the carrier of the remagnetization in the Trenton and Black River Formations.

The relationship between the formation of authigenic magnetite and the presence/migration of hydrocarbon has not yet been clearly established. The Michigan Basin is one of the best locations in North America to test the relationship because the Michigan Basin contains a well known, economically important oil/gas field, whose producing horizons (the Trenton and Black River Formations) are verified in this study to have been remagnetized during the late Paleozoic Kiaman Superchron. Authigenesis of magnetite during hydrocarbon migration or subsequent seepage has been suggested by Donovan et al. (1979), Elmore et al. (1987), and McCabe et al. (1987). For example, Donovan et al. (1979) reported that the location of aeromagnetic anomalies over the producing horizons at the Cement oil field, Oklahoma, indicates positive relation between formation of authigenic magnetite

and seepage of either petroleum or methane through strata overlying producing units. They hypothesized that magnetite formed by reduction of hematite and iron oxyhydroxides due to hydrocarbon-related brines. Reynolds et al. (1990), however, argued that the magnetite found by Donovan et al. (1979) was derived through contamination, possibly from drilling, because the magnetite has textures that resembles those of synthetic magnetite.

In this study, two basic patterns are observed. Within carbonate-rich intervals, little correlation is observed between the intensities of X_{arm} and carbon isotope excursions due to the attenuation of $-\delta^{13}C$ anomalies associated with hydrocarbon oxidation by the abundance of primary carbonate present in the bulk samples analyzed. In contrast, shaley intervals, poor in primary carbonate, exhibit a stronger overall correlation. Second, the overall X_{arm} intensities of producing, hyrocarbon-saturated strata in the producing well are smaller than those in the dry well. X_{arm} intensities in the carbonate and shale layers above the oil saturated strata in the producing well are considerably smaller than those in the same horizons in the dry well, through which hydrocarbons have migrated, but in which they are not now present.

It is important to note that magnetite authigenesis and formation of $-\delta^{13}C$ carbonate cements requires the presence of hydrocarbons within an oxidative diagenetic system, and a simple intermixture of hydrocarbons and subsurface fluids is not adequate to produce isotopically anomalous bicarbonate in the solution. Hydrocarbons must undergo oxidation to produce the HCO_3^- necessary for isotopic exchange with the fluid and for carbonate precipitation. Similarly, the formation of magnetite cannot occur under highly oxidizing or reducing conditions. In a subsurface hydrologic setting, this implies that magnetite authigenesis likely occurs at sites where hydrocarbons intermix with oxidizing fluids. Thus, magnetite authigenesis and increased X_{arm} intensities should not occur in horizons saturated with petroleum, but would likely develop at the interface between oil and water. The increased X_{arm} intensities of Ester Weir 1-5 well, compared to those in the Conklin 1-31 well, at and above the Trenton and Black River carbonates, which are laterally producing horizons, suggests enhanced magnetite authigenesis in strata through which hydrocarbons have migrated but in which they have not accumulated. In contrast to the producing horizons where hydrocarbons have saturated available porosity, interaction between migrating hydrocarbons and associated brines in the dry well (Ester Weir 1-5) could maintain the chemical setting necessary for carbon precipitation and magnetite formation during the migration of hydrocarbons. This process is similar to that suggested for methane seepage above hydrocarbon-producing fields, where methane is oxidized by shallow subsurface fluids. Such a mechanism has been envisioned for increased magnetite concentrations above the Cement oil field (Donovan et al. 1979) and could explain the increased X_{arm} intensities and $-\delta^{13}C$ anomalies present in the Coldwater Shale which overlies hydrocarbon production in Conklin 1-31 well. It is not clear, however, whether the magnetite authigenesis observed in the Coldwater Shale is related to the process of regional remagnetization, since paleomagnetic results for these younger rocks are not available. Although the $X_{arm} - \delta^{13}C$ results do not prove explicitly that the presence of hydrocarbons promotes formation of authigenic magnetite responsible for the late Paleozoic remagnetization, such results suggest the magnetite/hydrocarbon relationship more likely records processes associated with hydrocarbon migration in the strata through which they have passed.

The source of fluids responsible for regional remagnetization of Michigan Basin carbonates is not likely to be directly related to the tectonically expelled fluids from the Appalachian Orogenic belt (Oliver 1986; Dorobek 1989). The Michigan Basin is an isolated intracontinental basin bounded by several arches that probably restricted the influx of tectonically driven fluids from outside of the basin. This was evidenced by the fact that a hydrologic system in the Illinois Basin was found to be different from that of the Michigan Basin (Bethke 1986). A source of the fluids responsible for the remagnetization in the Trenton and Black River Formations is inferred to be a basinal brine mixed with meteoric recharge on the margins of the Michigan Basin, as suggested by Budai and Wilson (1991).

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